BROWARD COLLEGE

FPL Project Presentation

Correlation Analysis and Simple Linear Regression Models

By Guillermo Bonilla

Primary Objective

- The primary objective of this study is to examine how weather variables influence the electrical demand within Florida Power & Light's (FPL) service territory.

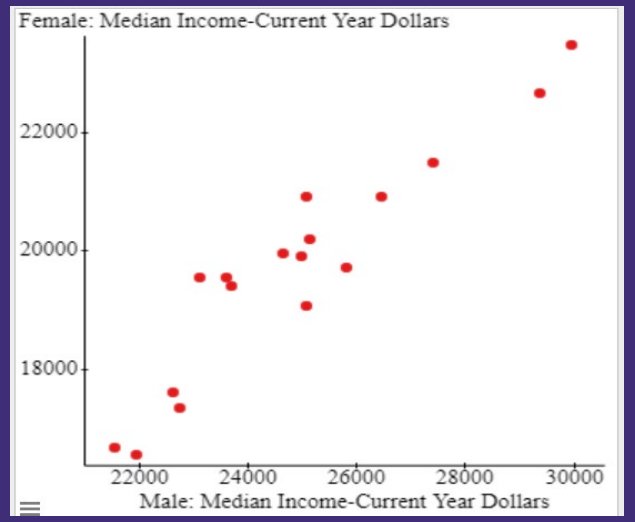
- By establishing a statistical relationship or correlation between meteorological factors and power usage, the project seeks to enhance forecasting accuracy and operational planning, particularly in an era marked by shifting climate patterns.

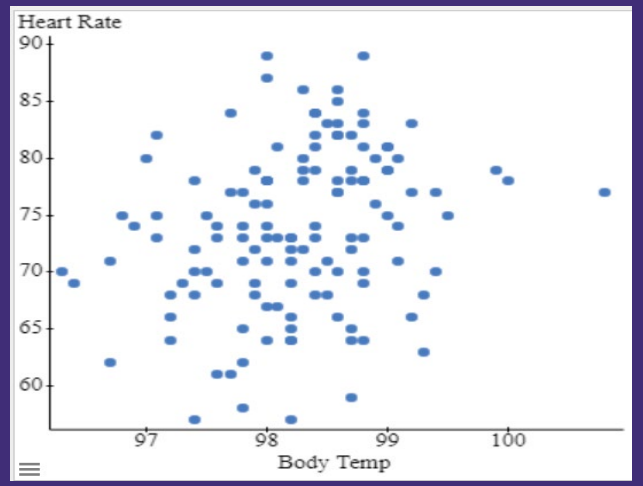
- This study will work with two variables at a time, using correlation analysis and simple linear regression models.

Correlation Analysis

- Correlation Analysis is a statistical technique used to analyze relationships between variables. It helps researchers to identify patterns and trends in data and make predictions based on the observed relationships.
- A correlation exists between two variables when the values of one variable are somehow associated with the values of the other variables.

Visualizing Correlation: Scatter Plots





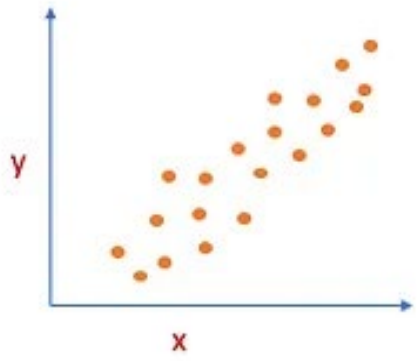
A **Scatter Plot** is a graphical representation used to display the relationship between two quantitative (numerical) variables, X and Y.

- It consists of points (x , y). In a scatter plot, each data point represents a single observation or measurement for both variables.
- Scatter plots are very useful for visually identifying patterns, trends, and relationships between variables.

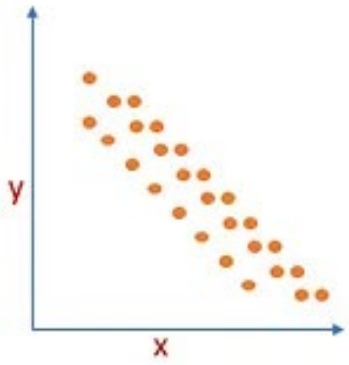
Linear Correlation

A **linear correlation** exists between two variables when the plotted points of paired data result in a pattern that can be approximated by a straight line.

A **positive** correlation means that as one variable increases, the other variable also increases.



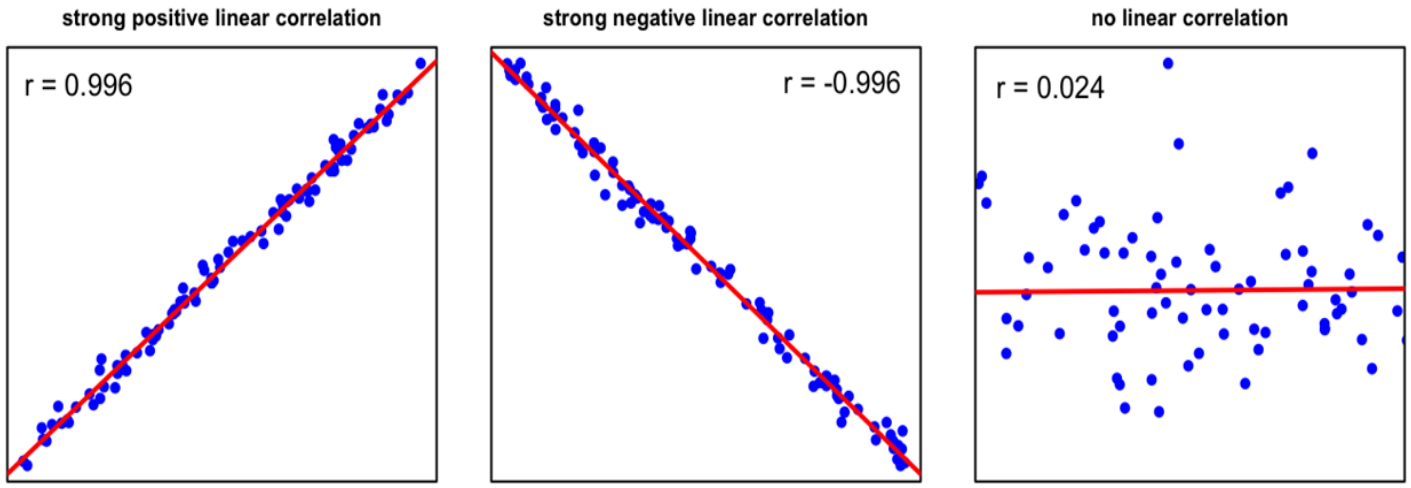
Positive Correlation



Negative Correlation

A **negative** correlation means that as one variable increases, the other variable decreases.

Pearson's Correlation Coefficient and Scatter plots



Linear correlation is measured using **Pearson's correlation coefficient (r)**, which quantifies the strength and direction of the **linear** relationship between two quantitative variables.

- If r is close to 1, there is a strong positive linear correlation between the two variables.
- If r is close to -1, there is a strong negative linear correlation between the two variables.
- If r is close to 0, there is no linear correlation between the two variables.

- Scatter plots may include a trend line or best-fit line that summarizes the overall pattern or trend in the data.
- This line is determined using a statistical method called Simple Linear Regression and can be used to make predictions.

My contribution to the project

- My contribution to the project involved analyzing the correlation between meteorological variables such as temperature and precipitation, and electrical demand and net generation.

- In cases where the correlation between the variables was statistically significant, simple linear regression models were obtained.

Simple Linear Regression

A Simple Linear Regression Model predicts a dependent variable Y by fitting a straight line to the data based on a single independent variable X. In essence, it produces a two-dimensional relationship, a linear equation, defined by one slope and one intercept.

$$Y_i = b_0 + b_1 X_{1i} + u_i$$

b_0 is the intercept

$b_1 X_{1i}$ is the slope

u_i is the error term

Dependent Variables	Independent Variables
Electrical Demand	Mean Temperature
Net Generation	Precipitation

In simple linear regression, R-squared, known as the coefficient of determination, is a statistical measure that indicates how well the regression model fits the observed data. R-squared values range from 0 to 1. If the value of R-squared is close to one, then the regression model is adequate, meaning there's substantial agreement between the proposed model and actual real-world observations. If the value of R-squared is close to zero, then the regression model is inadequate.

Electrical Demand and Net Generation Data

- The electrical demand data used in this study were sourced from the U.S. Energy Information Administration (EIA) database focusing on FPL data.

- Link:
  Real-time Operating Grid - U.S. Energy Information Administration (EIA)

- This comprehensive dataset comprises a daily time series capturing several key variables. The datetime variable records the specific day at which measurements were taken, while the demand (measured in MWh) represents the actual electricity consumption for each recorded day. Additionally, the dataset includes forecasted demand values, which facilitate a direct comparison between predicted and actual load, as well as metrics for net generation (MWh).

Meteorological Data

- Meteorological observations were obtained from the Florida State University (FSU) Climate Center, which captures daily weather information from numerous stations across Florida.
- Link:
  Downloadable Data - Florida Climate Center
- Each record in the dataset is associated with a specific weather station and an observation date, formed by merging the separate YEAR, MONTH, and DAY fields into a logical date format.
- The dataset includes daily measurements of precipitation, recorded in either inches or millimeters, along with temperature metrics that capture the maximum, minimum, and average daily temperatures.

Data Coverage

- Using the data from both sources, simple linear regression models were created to showcase a positive linear correlation between demand and mean temperature in the years 2019, 2020, 2021, 2022, 2023, and 2024 found in ten separate Florida locations (e.g., Ft. Lauderdale, Hialeah, Melbourne, Miami, Daytona Beach, Miami Beach, Vero Beach, Naples, Kissimmee, and Ft. Lauderdale Beach.)

Data Coverage Cont.

I obtained two separate models for each year and each location using the following dependent variables.

Y → Daily Electrical Demand

Y → Daily Net Generation

For both models the independent variable used was:

X →Daily Mean Temperature

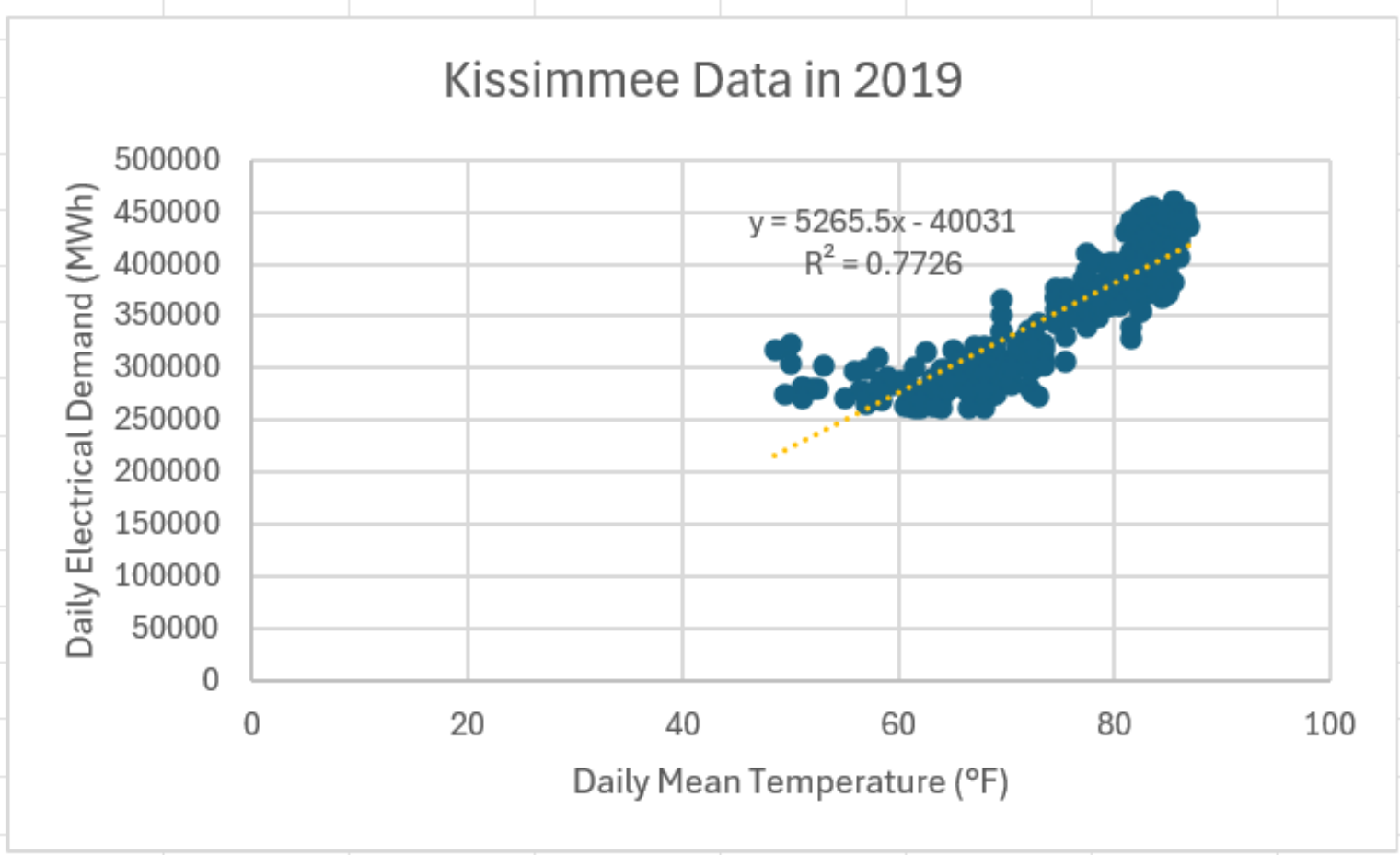
X → Daily Precipitation

For the purposes of the presentation, I will show results from the locations with the highest Pearson's correlation coefficients and R-squared values for the years 2019 and 2024.

Methodology

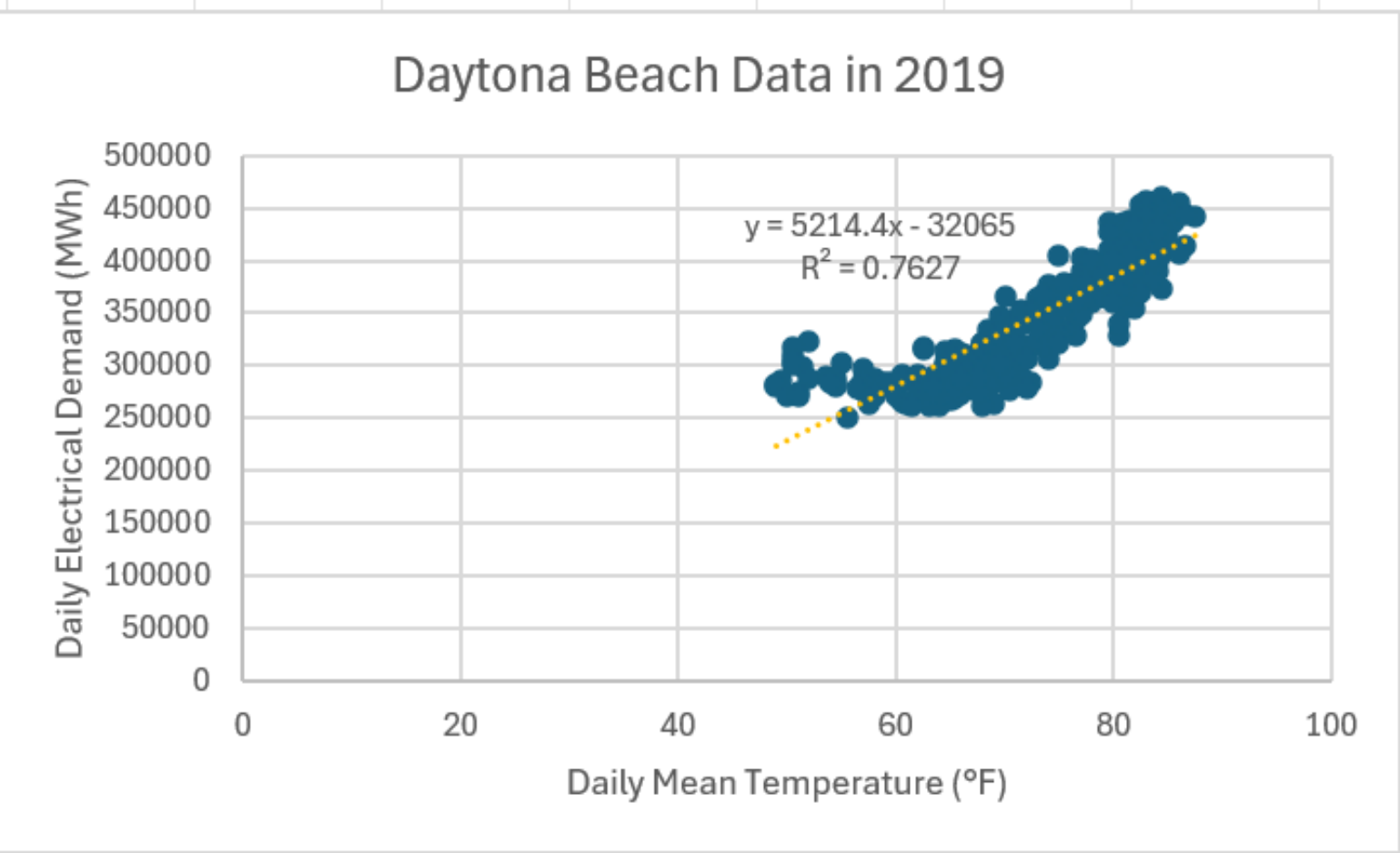
- The technology used to obtain the results was Excel spreadsheets. This included the statistical analysis tool packs.

Kissimmee 2019 Electrical Demand vs Mean Temperature Simple Linear Regression Model



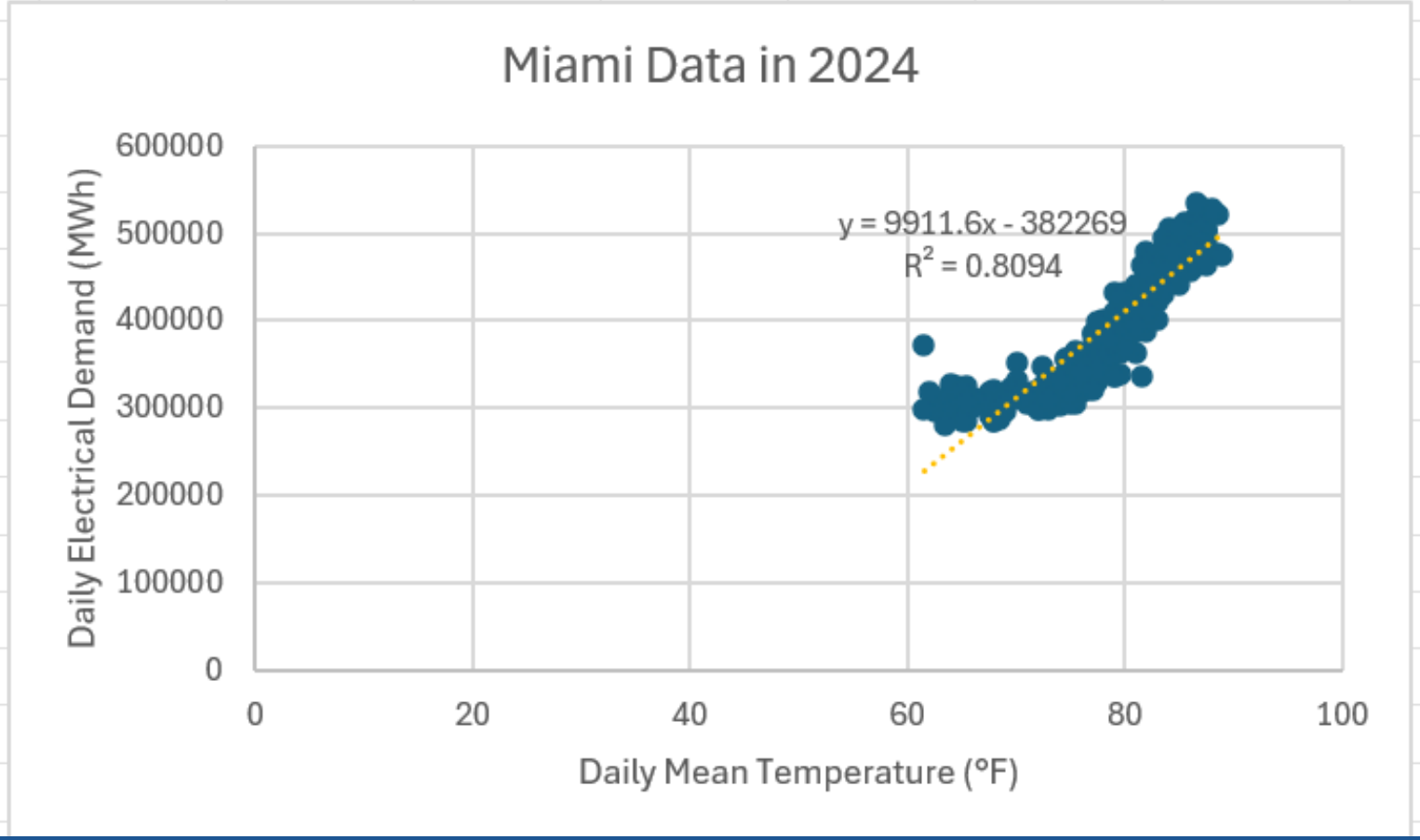
- A set of 313 observations were used to find the regression equation.
- The Pearson correlation coefficient between these two variables is 0.8790.
- A linear regression model is shown here between the daily electrical demand and daily mean temperature in Kissimmee in 2019. The equation of the line is
- $y = 5265.5x - 40031$,
- and the R squared value is 0.7726.

Daytona Beach 2019 Electrical Demand vs Mean Temperature Simple Linear Regression Model



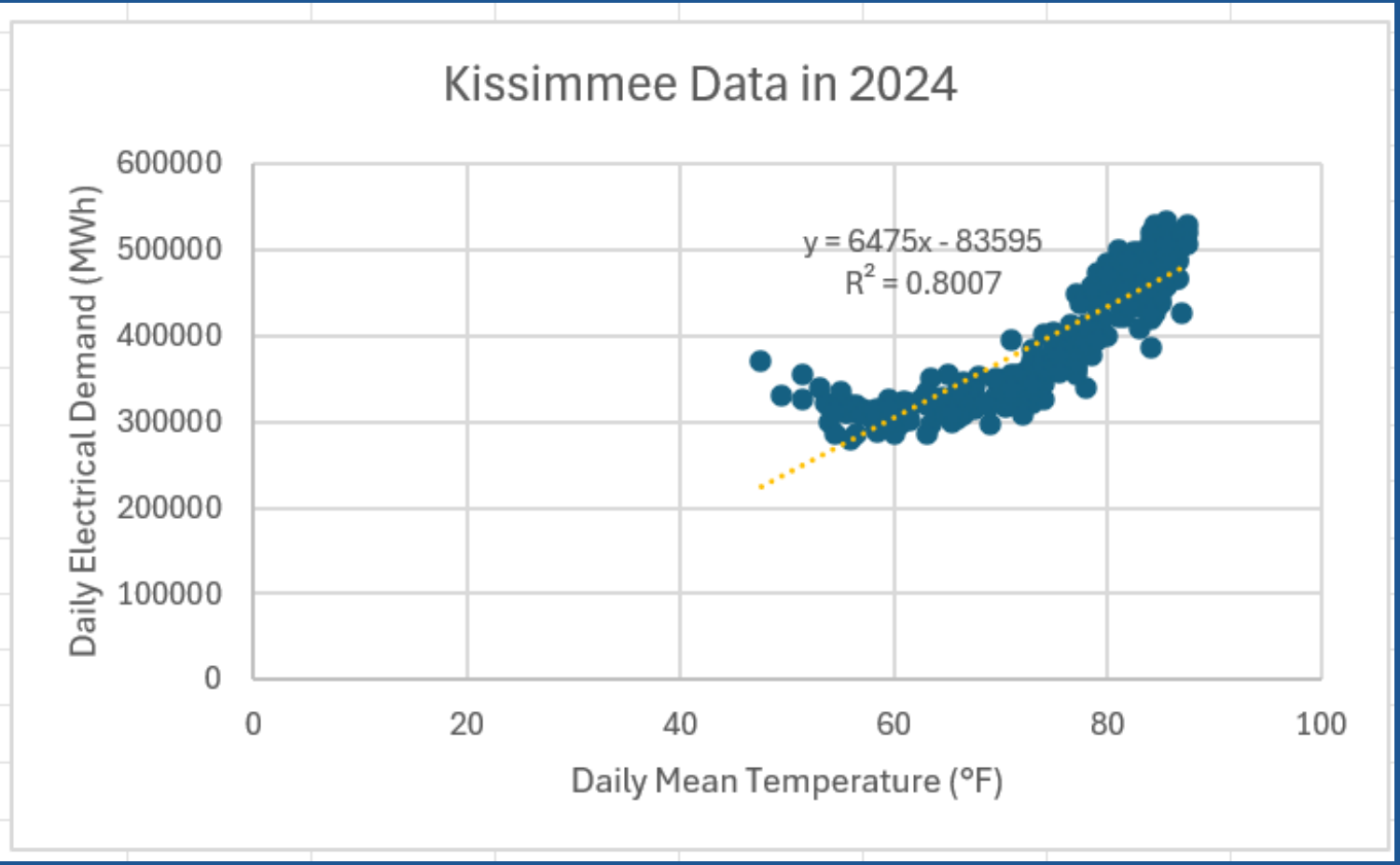
- A set of 359 observations were used to find the regression equation.
- The Pearson correlation coefficient between these two variables is 0.8733.
- A linear regression model is shown here between the daily electrical demand and daily mean temperature in Daytona Beach in 2019. The equation of the line is
- $y = 5214.4x - 32065$,
- and the R squared value is 0.7627.

Miami 2024 Electrical Demand vs Mean Temperature Simple Linear Regression Model



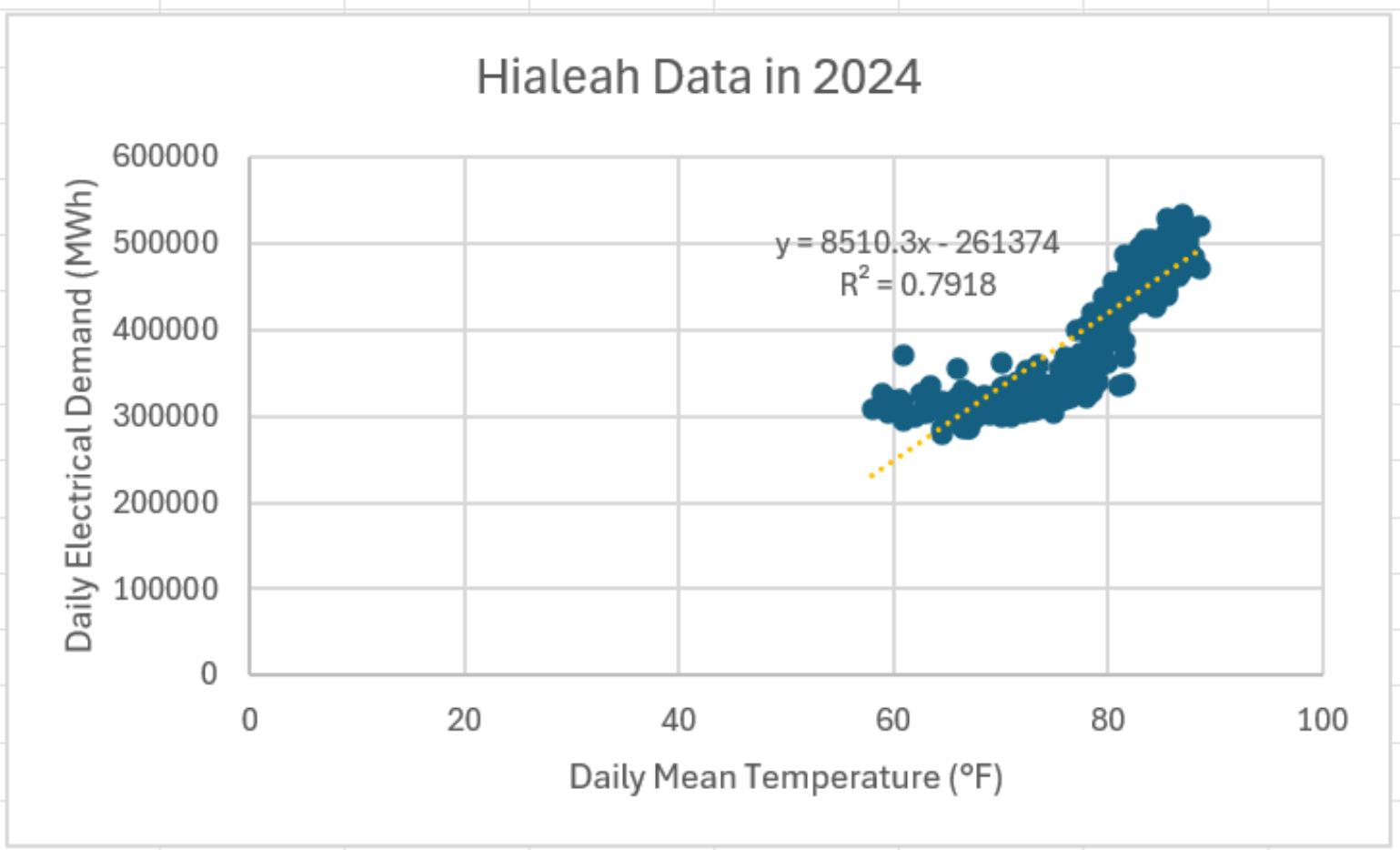
- A set of 366 observations were used to find the regression equation.
- The Pearson correlation coefficient between these two variables is 0.8997.
- A linear regression model is shown here between the daily electrical demand and daily mean temperature in Miami in 2024. The equation of the line is
- $y = 9911.6x - 382269$,
- and the R squared value is 0.8094.

Kissimmee 2024 Electrical Demand vs Mean Temperature Simple Linear Regression Model



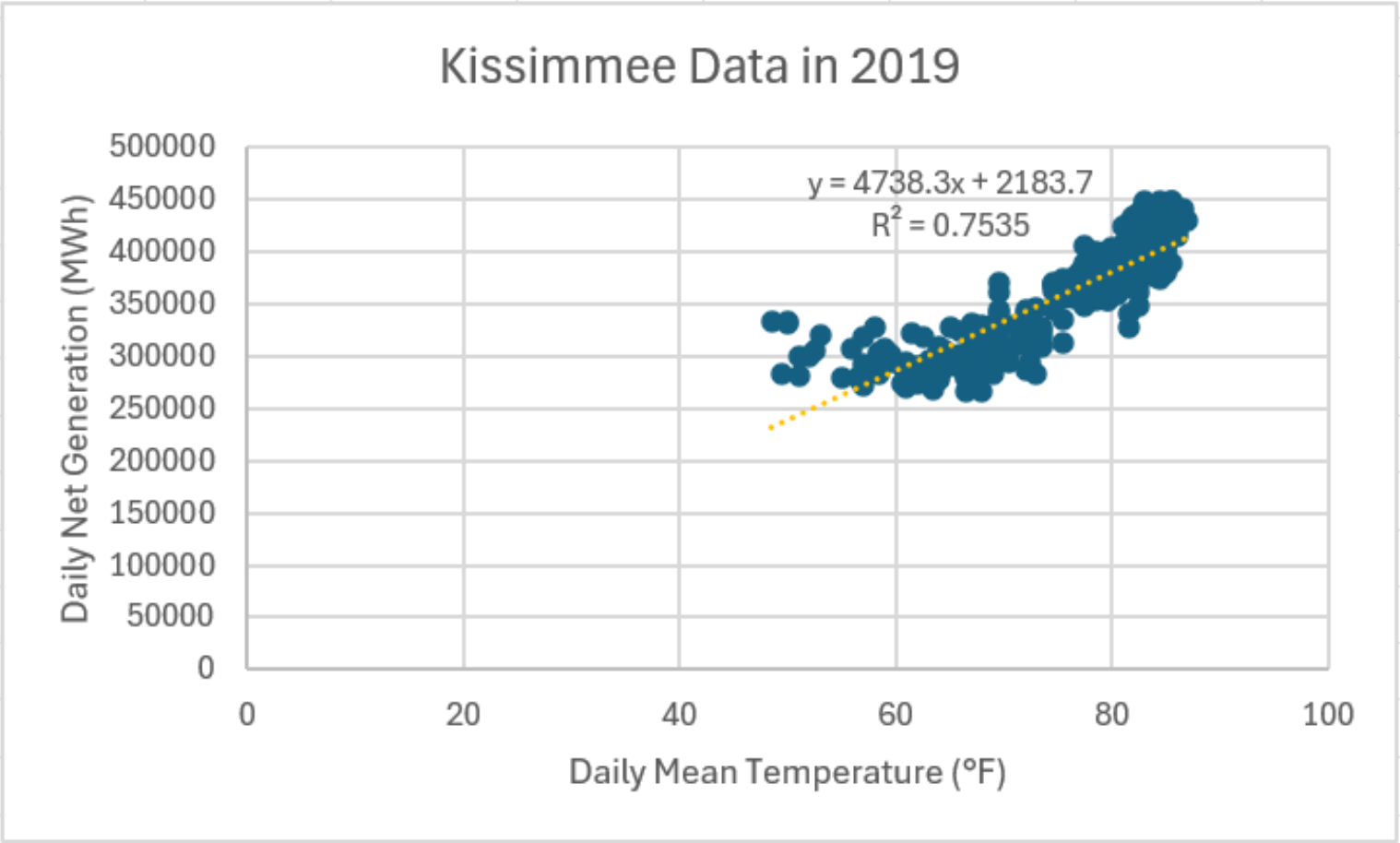
- A set of 330 observations were used to find the regression equation.
- The Pearson correlation coefficient between these two variables is 0.8948.
- A linear regression model is shown here between the daily electrical demand and daily mean temperature in Kissimmee in 2024. The equation of the line is
- $y = 6475x - 83595$,
- and the R squared value is 0.8007.

Hialeah 2024 Electrical Demand vs Mean Temperature Simple Linear Regression Model



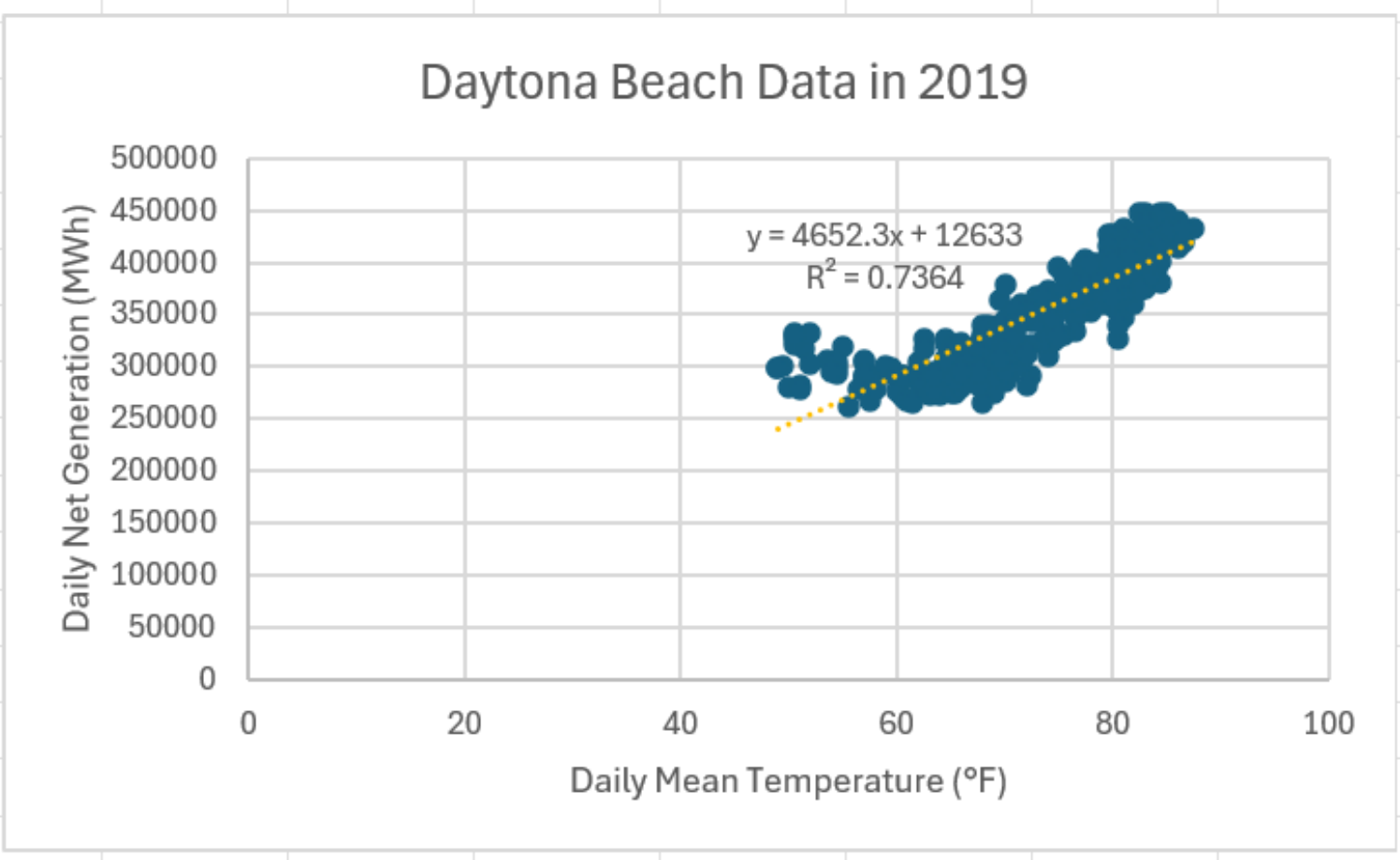
- A set of 324 observations were used to find the regression equation.
- The Pearson correlation coefficient between these two variables is 0.8897.
- A linear regression model is shown here between the daily electrical demand and daily mean temperature in Hialeah in 2024. The equation of the line is
- $y = 8510.3x - 261374$,
- and the R squared value is 0.7918.

Kissimmee 2019 Net Generation vs Mean Temperature Simple Linear Regression Model



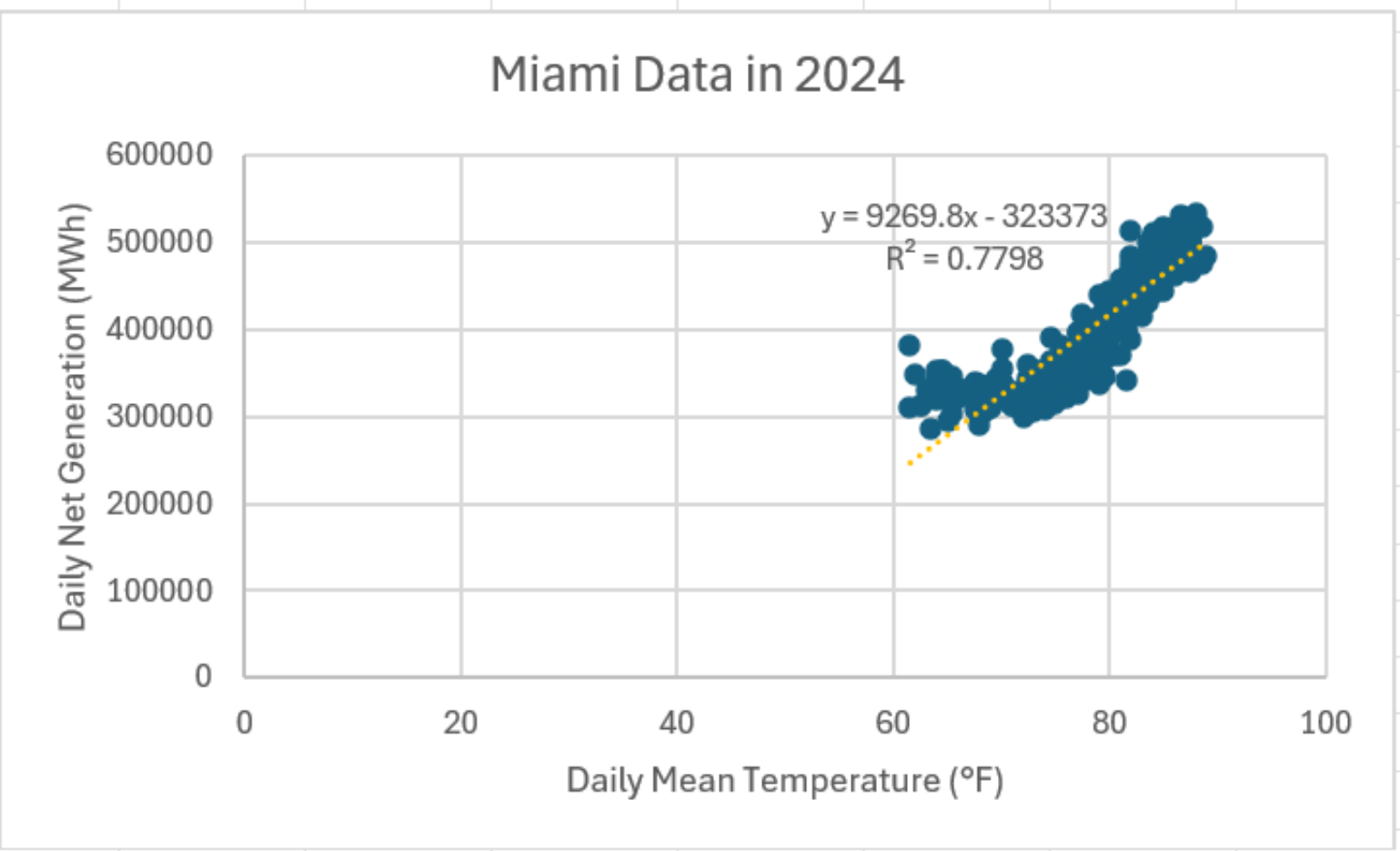
- A set of 313 observations were used to find the regression equation.
- The Pearson correlation coefficient between these two variables is 0.8680.
- A linear regression model is shown here between the daily net generation and daily mean temperature in Kissimmee in 2019. The equation of the line is
- $y = 4738.3x - 2183.7$,
- and the R squared value is 0.7535.

Daytona Beach 2019 Net Generation vs Mean Temperature Simple Linear Regression Model



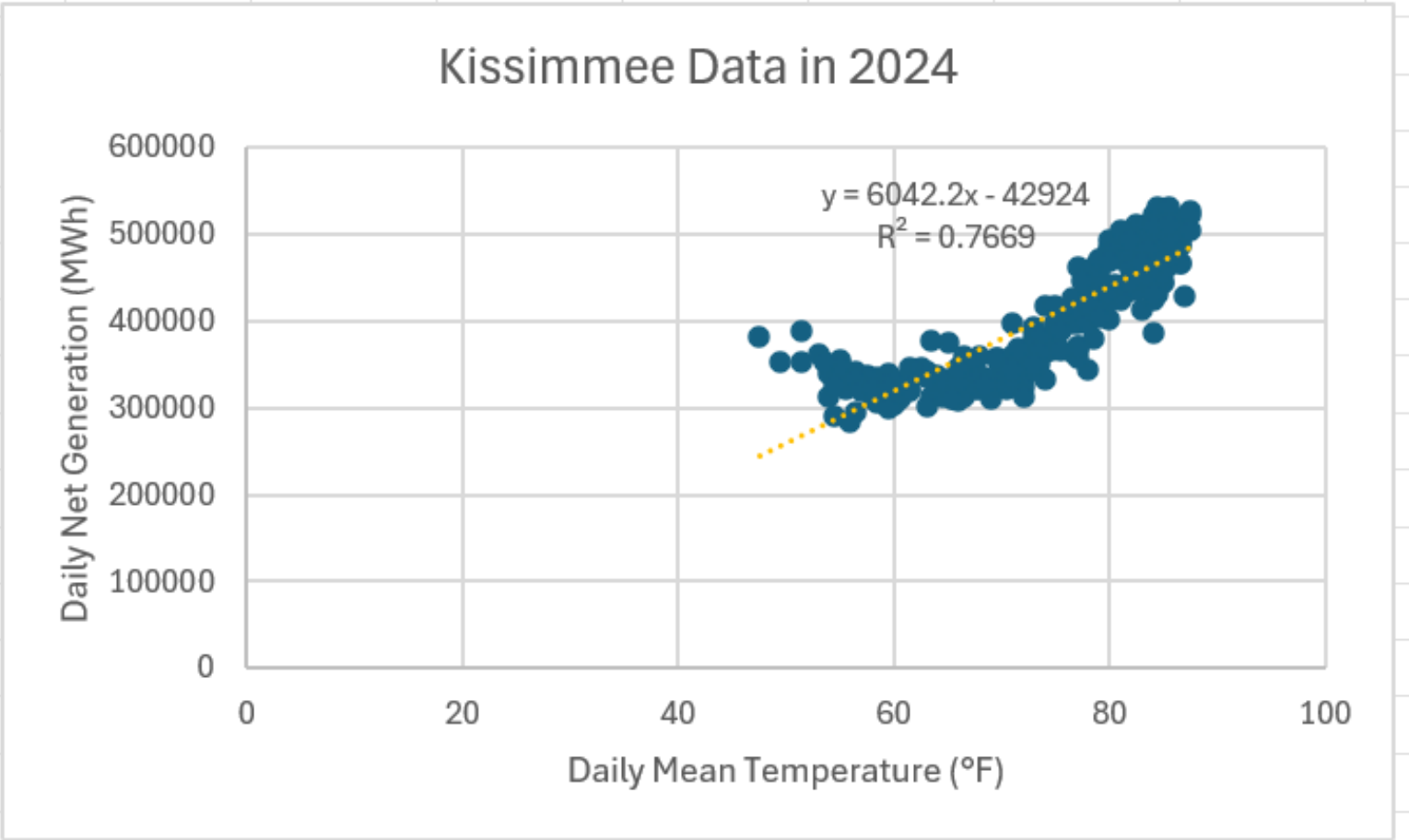
- A set of 359 observations were used to find the regression equation.
- The Pearson correlation coefficient between these two variables is 0.8581.
- A linear regression model is shown here between the daily net generation and daily mean temperature in Daytona Beach in 2019. The equation of the line is
- $y = 4652.3x + 12633$,
- and the R squared value is 0.7364.

Miami 2024 Net Generation vs Mean Temperature Simple Linear Regression Model



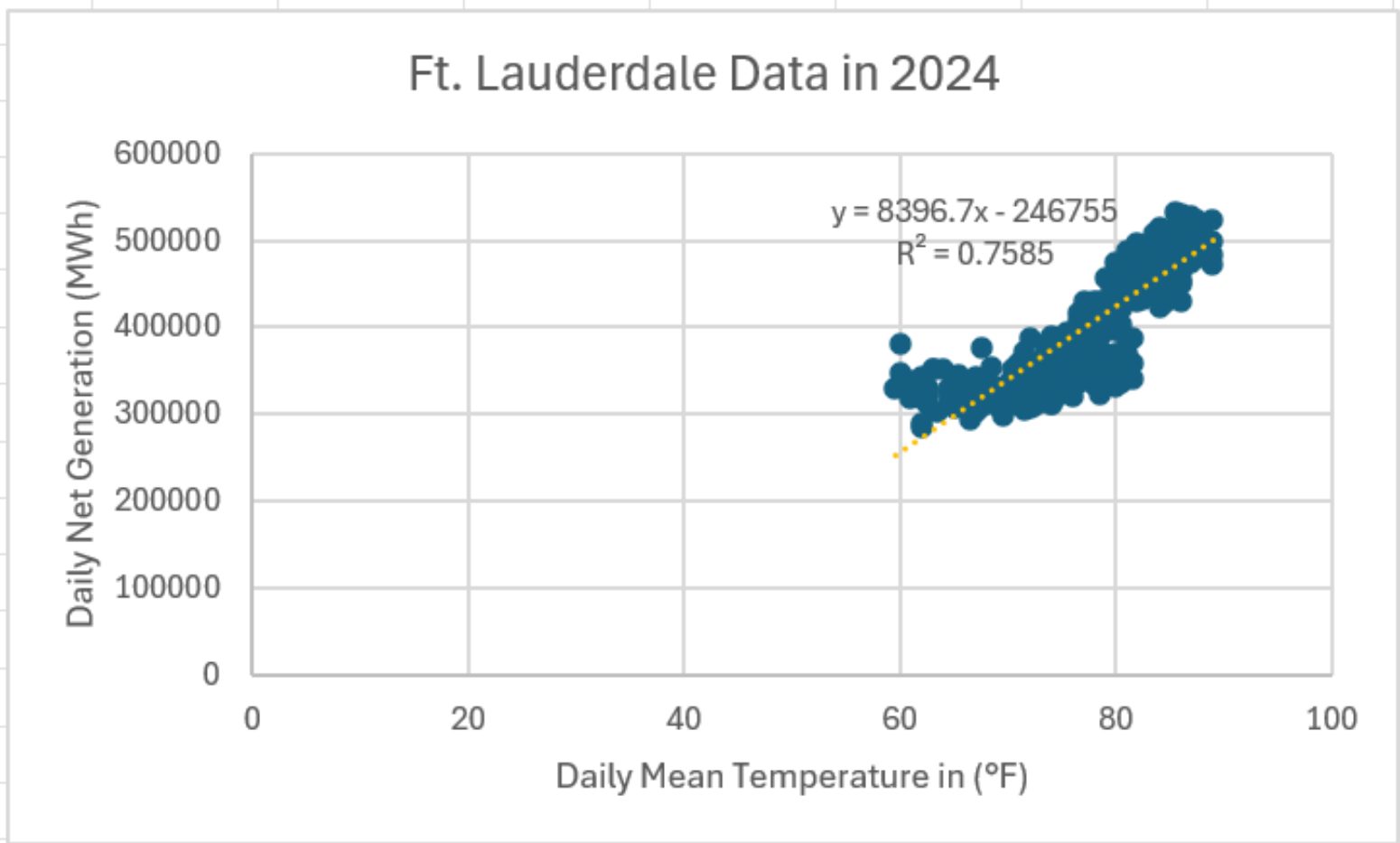
- A set of 366 observations were used to find the regression equation.
- The Pearson correlation coefficient between these two variables is 0.8831.
- A linear regression model is shown here between the daily net generation and daily mean temperature in Miami in 2024. The equation of the line is
- $y = 9269.8x - 323373$,
- and the R squared value is 0.7798.

Kissimmee 2024 Net Generation vs Mean Temperature Simple Linear Regression Model



- A set of 330 observations were used to find the regression equation.
- The Pearson correlation coefficient between these two variables is 0.8757.
- A linear regression model is shown here between the daily net generation and daily mean temperature in Kissimmee in 2024. The equation of the line is
- $y = 6042.2x - 42924$,
- and the R squared value is 0.7669.

Ft. Lauderdale 2024 Net Generation vs Mean Temperature Simple Linear Regression Model

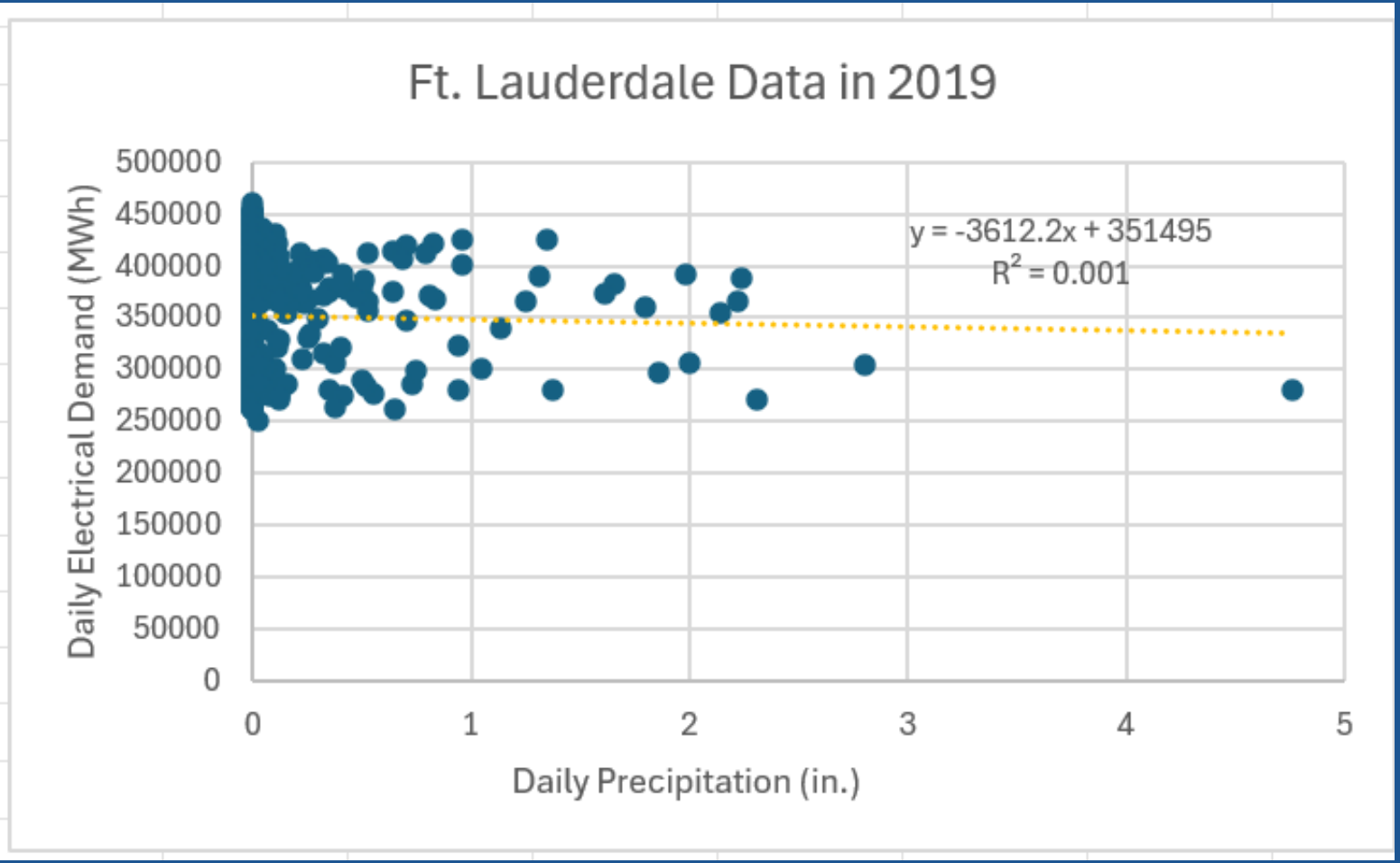


- A set of 364 observations were used to find the regression equation.
- The Pearson correlation coefficient between these two variables is 0.8709.
- A linear regression model is shown here between the daily net generation and daily mean temperature in Ft. Lauderdale in 2024. The equation of the line is
- $y = 8396.7x - 246755$,
- and the R squared value is 0.7585.

Additional Findings

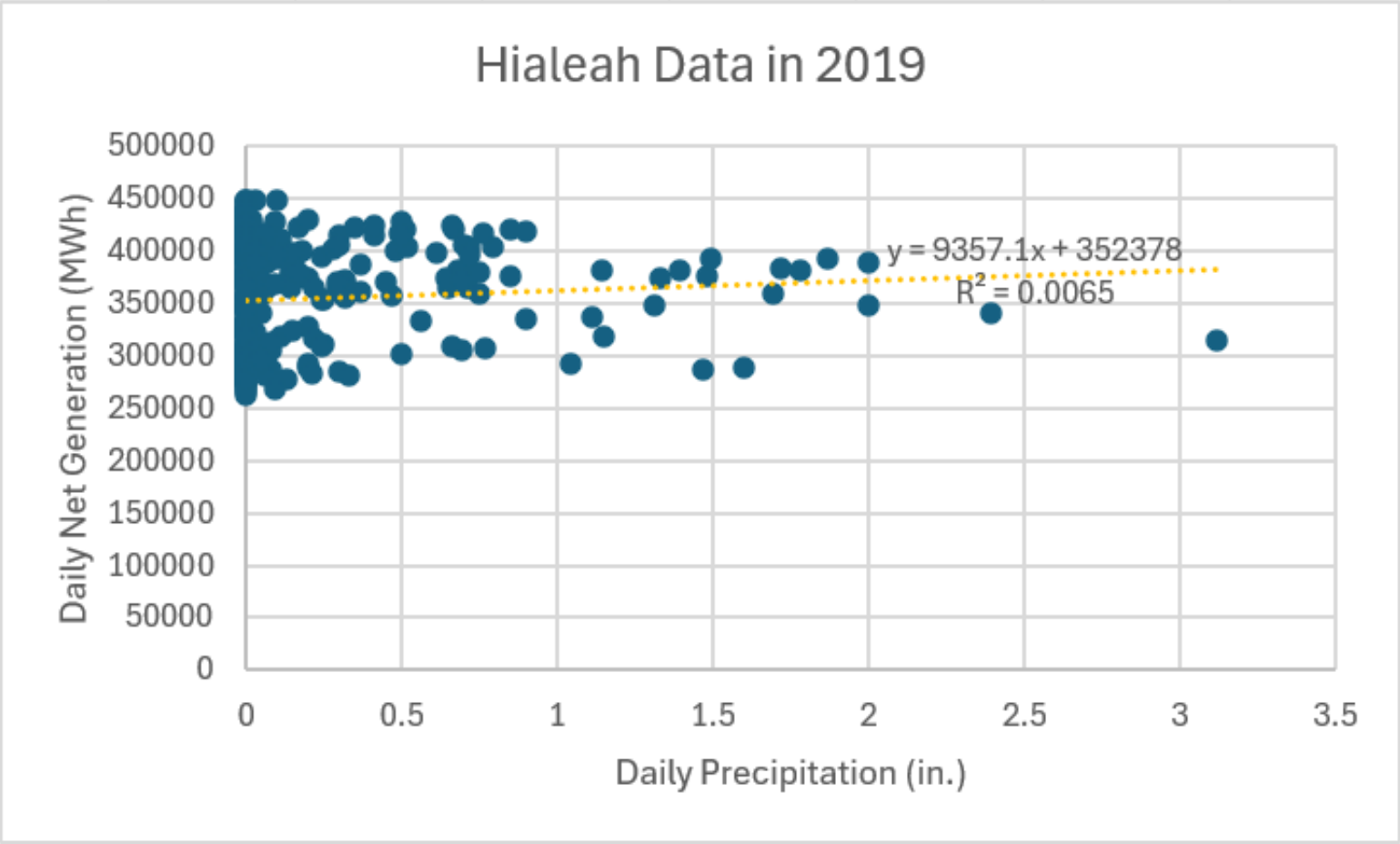
- The independent variable precipitation was also analyzed briefly in separate models across demand and net generation.
- These models overall showed weak linear correlation. Moreover, very weak R-squared values and weak Pearson's correlations coefficients were outputted as well.
- This is shown in the following graphs.

Electrical Demand vs Precipitation Example



- A set of 359 observations were used to find the regression equation.
- The Pearson correlation coefficient between these two variables is -0.0316.
- A linear regression model is shown here between the daily electrical demand and daily precipitation in Ft. Lauderdale in 2019. The equation of the line is
- $y = -3612.2x - 351495$,
- and the R squared value is 0.001.

Net Generation vs Precipitation Example



A set of 332 observations were used to find the regression equation.

The Pearson correlation coefficient between these two variables is 0.0806.

A linear regression model is shown here between the daily net generation and daily precipitation in Hialeah in 2019. The equation of the line is

$y = 9357.1x + 352378,$

and the R squared value is 0.0065.

Main Conclusions

•According to the statistical outputs, the results show there is a strong positive linear correlation between mean temperature and electrical demand, and between mean temperature and net generation.

• There is also evidence for more electrical consumption when temperatures rises in each of the ten Florida locations.

• Conversely, there is less electrical consumption when temperature is lower in each of the ten cities.

• Also, by including more variables in the linear regression model will increase the goodness of fit indicator or R-squared coefficient. In the next presentation, this will be further discussed.

•In addition, the results show that cities comprised of much tourism and coastal areas and higher population exhibit stronger positive linear correlation due to the higher R-squared and Pearson's correlation coefficient values.

References

- Elementary Statistics 14th edition
  Mario F. Triola Chapter 10 Correlation and Regression
  U.S. Energy Information Administration (EIA) database focusing on FPL data
  Link:

  Real-time Operating Grid - U.S. Energy Information Administration (EIA)
- Meteorological observations were obtained from the Florida State University (FSU) Climate Center, which captures daily weather information from numerous stations across Florida.
  Link:

  Downloadable Data - Florida Climate Center

Questions?